Reg. Number:	22BAI1268

VIT
Vellore Institute of Technology
CHENNAI

Continuous Assessment Test (CAT) – II - MARCH 2025

Programme	:	B.Tech.CSE and its Specializations	Semester	:	Winter 2024-25
Course Code & Course Title	:	BCSE309L & Cryptography and Network Security	Class Number	:	CH2024250502355 CH2024250502351 CH2024250502352 CH2024250502346 CH2024250502349 CH2024250501876 CH2024250501874 CH2024250502668
Faculty	:	Dr. Renuka Devi S Dr. Anita X Dr. Rajesh R Dr. Jannath Nisha O S Dr. Tapabatra Roy Dr. Kanthimathi S Dr. Logeshwari G Prof. Jai Vinita L	Slot	:	F1+TF1
Duration	:	**1 ½ Hours**	Max. Mark		50

Answer all questions

1		A multinational bank wants to encrypt customer transactions using IDEA. Explain how IDEA would be implemented for secure data transmission. Discuss the encryption and decryption process, key management, and security considerations.	10
2		Choose the Elliptic curve with prime $p = 13$, $a = 2$, $b = 4$. Base point $G = (5,10)$ and message $M = (5,3)$. The receiver's private key is 2, and the sender chooses a random integer $k = 3$. Compute the cipher text.	10
3		Mr. Arjun, Mr. Bhavesh, and Mr. Chirag would like to initiate a secret chat between them for the cybersecurity project as they are working on for ISRO. They need a common secret key that can be used in the ChaCha20 encryption algorithm for their secure communication. Help them generate the common secret key using the Diffie-Hellman algorithm with the following parameters: prime $Q = 19$, $\alpha = 3$ primitive root of Q, the private keys of Mr. Arjun, Mr. Bhavesh, and Mr. Chirag are 4, 5, and 6, respectively. Determine the common secret key that will be used for ChaCha20 encryption.	10

4	i.	A cybersecurity company is implementing SHA-512 to securely hash user passwords. A developer is testing the hashing process with different plaintext messages. Consider the following: Message: 1 of 300 bytes. Message: 2 of 127 bytes. a) Determine the bit length of each message before padding. (1 mark) b) Calculate the number of padding bits required for each message before appending original length of the message.(2 marks) c) Determine the number of message blocks to be processed in each message for generating final hash value.(2 marks)	10
	ii.	A company is implementing HMAC-MD5 for message authentication between a client and a server. The client wants to send a message securely, ensuring integrity and authenticity using a shared secret key. Secret key (K): 0x 4D 65 73 73 Compute Inner Padding (S_i) with the ipad constant 0x36. (5 marks)	
5		Assume that you have received a message along with ElGamal based digital signature (17, 2) from your business partner. Given the hash of the message is 15 and the public key of the sender is (23, 5, 9). a) Verify the signature and comment on it (3 marks) b) Show the stepwise process of signature generation and verification with the given data assuming the private key of the sender is (23, 5, 10) and the random value K is 7. (7 marks)	10

**********All the best **********